Hierarchical Graph Swarm Learning for Personalised and Energy-Efficient Edge Intelligence

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APPENDIX

A. Conditional Convergence Bounds under Time-Varying Communication Graphs

This section presents the conditional convergence analysis of the idealized optimization dynamics and the equilibrium-existence analysis of the game-theoretic formulations in HierGSL. We first derive optimization bounds under time-varying communication graphs and then establish the existence of the corresponding game equilibria. The convergence of the MADDPG and IDQN policy-learning processes themselves is not claimed.

The proposed HierGSL framework adaptively determines the intra-cluster and inter-cluster communication relationships through MADDPG and IDQN, respectively. Let

$$\mathbf{W}(t) = \mathbf{W}_{\text{inter}}(t)\mathbf{W}_{\text{intra}}(t) \tag{1}$$

denote the effective aggregation matrix at communication round t. The optimization component of HierGSL is represented as

$$\mathbf{\Theta}(t+1) = \mathbf{W}(t)\left[\mathbf{\Theta}(t) - \alpha\mathbf{G}(t)\right], \tag{2}$$

where

$$\mathbf{\Theta}(t) = \begin{bmatrix} \theta_1(t)^T \\ \vdots \\ \theta_N(t)^T \end{bmatrix} \in \mathbb{R}^{N\times d}$$

collects the local models,

$$\mathbf{G}(t) = \begin{bmatrix} \nabla F_1(\theta_1(t))^T \\ \vdots \\ \nabla F_N(\theta_N(t))^T \end{bmatrix}$$

collects the local gradients, and $\alpha > 0$ denotes the local optimization step size.

The following analysis is conditional on the realized sequence of effective aggregation matrices. It characterizes the optimization process for a graph sequence satisfying the assumptions below and does not analyze the convergence of the MADDPG or IDQN policy-learning processes.

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Assumption 1 (Smooth local objectives). Each local objective $F_i : \mathbb{R}^d \to \mathbb{R}$ is L-smooth:

$$\|\nabla F_i(\theta) - \nabla F_i(\theta')\| \le L\|\theta - \theta'\|, \qquad \forall \theta, \theta' \in \mathbb{R}^d. \tag{3}$$

The aggregate objective

$$F(\theta) = \frac{1}{N}\sum_{i=1}^{N} F_i(\theta) \tag{4}$$

is lower bounded by $F_{\text{inf}} > -\infty$.

Assumption 2 (Balanced effective aggregation). For every communication round t, $\mathbf{W}(t)$ is non-negative and doubly stochastic:

$$\mathbf{W}(t)\mathbf{1} = \mathbf{1}, \qquad \mathbf{1}^T\mathbf{W}(t) = \mathbf{1}^T. \tag{5}$$

This is an analytical condition on the effective matrix $\mathbf{W}(t)$ and is not implied solely by the use of MADDPG or IDQN.

Assumption 3 (Uniform mixing). Let

$$\mathbf{\Phi}(t,s) = \mathbf{W}(t)\mathbf{W}(t-1)\cdots\mathbf{W}(s), \qquad t \ge s, \tag{6}$$

and define

$$\mathbf{J} = \frac{1}{N}\mathbf{1}\mathbf{1}^T. \tag{7}$$

There exist constants $C > 0$ and $\rho \in (0,1)$ such that

$$\|\mathbf{\Phi}(t,s) - \mathbf{J}\|_2 \le C\rho^{t-s+1}, \qquad \forall t \ge s. \tag{8}$$

This condition can be obtained under standard uniform joint-connectivity, positive-self-weight, and lower-bounded-weight conditions for an appropriately balanced graph sequence. In the proposed analysis, it is imposed directly on the realized effective matrix sequence.

Assumption 4 (Bounded local-gradient disagreement). Along the generated iterates, the disagreement among the local gradients is uniformly bounded:

$$\|(\mathbf{I}_N - \mathbf{J})\,\mathbf{G}(t)\|_F \le G_\perp, \qquad \forall t, \tag{9}$$

for some finite constant $G_\perp > 0$.

Define the network-average model and the model-disagreement component as

$$\bar{\theta}(t) = \frac{1}{N}\mathbf{1}^T\mathbf{\Theta}(t), \tag{10}$$

and

$$\mathbf{\Theta}_\perp(t) = (\mathbf{I}_N - \mathbf{J})\,\mathbf{\Theta}(t), \tag{11}$$

respectively.

Lemma 1 (Disagreement bound)**.** Under Assumptions 2–4, the sequence generated by (2) satisfies

$$\|\mathbf{\Theta}_\perp(t)\|_F \le C\rho^t \|\mathbf{\Theta}_\perp(0)\|_F + \frac{C\alpha G_\perp}{1-\rho}. \tag{12}$$

Consequently,

$$\limsup_{t\to\infty} \|\mathbf{\Theta}_\perp(t)\|_F \le D_\alpha, \qquad D_\alpha \triangleq \frac{C\alpha G_\perp}{1-\rho}. \tag{13}$$

Proof. Using the double stochasticity of $\mathbf{W}(t)$, we have

$$\mathbf{J}\mathbf{W}(t) = \mathbf{W}(t)\mathbf{J} = \mathbf{J}.$$

Therefore,

$$\begin{aligned}\mathbf{\Theta}_\perp(t+1) &= (\mathbf{I}_N - \mathbf{J})\,\mathbf{W}(t)\,[\mathbf{\Theta}(t) - \alpha\mathbf{G}(t)] \\ &= (\mathbf{W}(t) - \mathbf{J})\,\mathbf{\Theta}_\perp(t) - \alpha\,(\mathbf{W}(t) - \mathbf{J})\,\mathbf{G}_\perp(t),\end{aligned} \tag{14}$$

where

$$\mathbf{G}_\perp(t) = (\mathbf{I}_N - \mathbf{J})\,\mathbf{G}(t).$$

Unrolling (14) gives

$$\begin{aligned}\mathbf{\Theta}_\perp(t) = {} & [\mathbf{\Phi}(t-1,0) - \mathbf{J}]\,\mathbf{\Theta}_\perp(0) \\ & - \alpha\sum_{s=0}^{t-1} [\mathbf{\Phi}(t-1,s) - \mathbf{J}]\,\mathbf{G}_\perp(s).\end{aligned} \tag{15}$$

Applying Assumptions 3 and 4 yields

$$\begin{aligned}\|\mathbf{\Theta}_\perp(t)\|_F &\le C\rho^t \|\mathbf{\Theta}_\perp(0)\|_F + C\alpha G_\perp \sum_{s=0}^{t-1} \rho^{t-s} \\ &\le C\rho^t \|\mathbf{\Theta}_\perp(0)\|_F + \frac{C\alpha G_\perp}{1-\rho}.\end{aligned} \tag{16}$$

This proves (12). □

The double stochasticity of $\mathbf{W}(t)$ further gives the correct network-average update

$$\begin{aligned}\bar\theta(t+1) &= \frac{1}{N}\mathbf{1}^T\mathbf{\Theta}(t+1) \\ &= \bar\theta(t) - \alpha\bar{\mathbf{g}}(t),\end{aligned} \tag{17}$$

where

$$\bar{\mathbf{g}}(t) = \frac{1}{N}\sum_{i=1}^N \nabla F_i(\theta_i(t))\,. \tag{18}$$

In general, $\bar{\mathbf{g}}(t) \neq \nabla F(\bar\theta(t))$. Define the corresponding gradient-mismatch term as

$$\mathbf{r}(t) = \bar{\mathbf{g}}(t) - \nabla F\big(\bar\theta(t)\big)\,. \tag{19}$$

By Assumption 1,

$$\begin{aligned}\|\mathbf{r}(t)\| &\le \frac{1}{N}\sum_{i=1}^N \big\|\nabla F_i(\theta_i(t)) - \nabla F_i(\bar\theta(t))\big\| \\ &\le \frac{L}{N}\sum_{i=1}^N \big\|\theta_i(t) - \bar\theta(t)\big\| \\ &\le \frac{L}{\sqrt{N}}\,\|\mathbf{\Theta}_\perp(t)\|_F\,.\end{aligned} \tag{20}$$

Hence, (17) is equivalently written as

$$\bar\theta(t+1) = \bar\theta(t) - \alpha\left[\nabla F\big(\bar\theta(t)\big) + \mathbf{r}(t)\right]. \tag{21}$$

Theorem 1 (Strongly convex objective)**.** Suppose that Assumptions 1–4 hold and that the aggregate objective $F(\theta)$ is additionally μ-strongly convex, with unique minimizer $\theta^\star$. If

$$0 < \alpha \le \frac{2}{L+\mu}, \tag{22}$$

then

$$\limsup_{t\to\infty} \big\|\bar\theta(t) - \theta^\star\big\| \le \frac{\alpha L D_\alpha}{\sqrt{N}(1-q)}, \tag{23}$$

where

$$q = \sqrt{1 - \frac{2\alpha\mu L}{\mu + L}} < 1. \tag{24}$$

Consequently,

$$\begin{aligned}&\limsup_{t\to\infty} \left(\frac{1}{N}\sum_{i=1}^N \|\theta_i(t) - \theta^\star\|^2\right)^{1/2} \\ &\quad\le \frac{D_\alpha}{\sqrt{N}}\left(1 + \frac{\alpha L}{1-q}\right) = O\!\left(\frac{\alpha}{1-\rho}\right).\end{aligned} \tag{25}$$

Proof. For an L-smooth and μ-strongly convex function, the centralized gradient mapping satisfies

$$\big\|\bar\theta(t) - \alpha\nabla F(\bar\theta(t)) - \theta^\star\big\| \le q\,\big\|\bar\theta(t) - \theta^\star\big\|\,. \tag{26}$$

Using (21), we obtain

$$\begin{aligned}\big\|\bar\theta(t+1) - \theta^\star\big\| &\le q\,\big\|\bar\theta(t) - \theta^\star\big\| + \alpha\,\|\mathbf{r}(t)\| \\ &\le q\,\big\|\bar\theta(t) - \theta^\star\big\| + \frac{\alpha L}{\sqrt{N}}\,\|\mathbf{\Theta}_\perp(t)\|_F\,.\end{aligned} \tag{27}$$

Taking the limit superior and using Lemma 1 gives

$$\limsup_{t\to\infty} \big\|\bar\theta(t) - \theta^\star\big\| \le q\limsup_{t\to\infty} \big\|\bar\theta(t) - \theta^\star\big\| + \frac{\alpha L D_\alpha}{\sqrt{N}},$$

which proves (23). Finally,

$$\begin{aligned}&\left(\frac{1}{N}\sum_{i=1}^N \|\theta_i(t) - \theta^\star\|^2\right)^{1/2} \\ &\le \frac{1}{\sqrt{N}}\,\|\mathbf{\Theta}_\perp(t)\|_F + \big\|\bar\theta(t) - \theta^\star\big\|\,,\end{aligned} \tag{28}$$

which yields (25). □

Theorem 1 does not claim exact convergence for a fixed step size. The residual reflects both persistent local-gradient disagreement and incomplete mixing over the adaptive communication graph.

Theorem 2 (Smooth non-convex objective)**.** Suppose that Assumptions 1–4 hold and that

$$0 < \alpha \le \frac{1}{4L}. \tag{29}$$

Then, for any $T \geq 1$,

$$\frac{1}{T}\sum_{t=0}^{T-1}\left\|\nabla F\big(\bar{\theta}(t)\big)\right\|^2 \leq \frac{2\left[F(\bar{\theta}(0)) - F_{\inf}\right]}{\alpha T} + \frac{5L^2}{2NT}\sum_{t=0}^{T-1}\left\|\mathbf{\Theta}_{\perp}(t)\right\|_F^2. \tag{30}$$

Consequently,

$$\limsup_{T\to\infty}\frac{1}{T}\sum_{t=0}^{T-1}\left\|\nabla F\big(\bar{\theta}(t)\big)\right\|^2 \leq \frac{5L^2D_\alpha^2}{2N} = O\left(\frac{\alpha^2}{(1-\rho)^2}\right). \tag{31}$$

Proof. For notational simplicity, let

$$\mathbf{g}(t) = \nabla F\big(\bar{\theta}(t)\big).$$

Using (21) and the smoothness of F,

$$F(\bar{\theta}(t+1)) \leq F(\bar{\theta}(t)) - \alpha\left\langle \mathbf{g}(t), \mathbf{g}(t) + \mathbf{r}(t)\right\rangle + \frac{L\alpha^2}{2}\left\|\mathbf{g}(t) + \mathbf{r}(t)\right\|^2. \tag{32}$$

Using

$$-\left\langle \mathbf{g}(t), \mathbf{r}(t)\right\rangle \leq \frac{1}{4}\|\mathbf{g}(t)\|^2 + \|\mathbf{r}(t)\|^2$$

and

$$\|\mathbf{g}(t) + \mathbf{r}(t)\|^2 \leq 2\|\mathbf{g}(t)\|^2 + 2\|\mathbf{r}(t)\|^2,$$

we obtain, for $\alpha \leq 1/(4L)$,

$$F(\bar{\theta}(t+1)) \leq F(\bar{\theta}(t)) - \frac{\alpha}{2}\|\mathbf{g}(t)\|^2 + \frac{5\alpha}{4}\|\mathbf{r}(t)\|^2. \tag{33}$$

Summing from $t = 0$ to $T-1$, using the lower boundedness of F, and applying (20) gives

$$\frac{1}{T}\sum_{t=0}^{T-1}\|\mathbf{g}(t)\|^2 \leq \frac{2[F(\bar{\theta}(0)) - F_{\inf}]}{\alpha T} + \frac{5L^2}{2NT}\sum_{t=0}^{T-1}\left\|\mathbf{\Theta}_{\perp}(t)\right\|_F^2. \tag{34}$$

Finally, Lemma 1 implies

$$\limsup_{t\to\infty}\left\|\mathbf{\Theta}_{\perp}(t)\right\|_F \leq D_\alpha,$$

which proves (31). □

Therefore, under a fixed optimization step size, the revised analysis establishes bounded disagreement and convergence to an optimization neighborhood. It does not claim exact consensus, exact convergence to the global minimizer, or exact convergence to a first-order stationary point. The sizes of the residual neighborhoods decrease as the optimization step size decreases and as the mixing quality of the realized graph sequence improves.

a) Empirical diagnostics of the learned aggregation matrices.: We examine the effective aggregation matrices recorded in the experiments, using the same executed intra- and inter-cluster weights for both checks, without additional normalization. The minimum weight is zero. The balance errors are defined as

$$\epsilon_{\mathrm{row}}(t) = \left\|\mathbf{W}(t)\mathbf{1} - \mathbf{1}\right\|_\infty,$$
$$\epsilon_{\mathrm{col}}(t) = \left\|\mathbf{W}(t)^T\mathbf{1} - \mathbf{1}\right\|_\infty.$$

Their maxima over the evaluated rounds are $1.11022302 \times 10^{-16}$ and $3.42029842 \times 10^{-16}$, respectively, supporting Assumption 2 to numerical precision for the recorded matrices.

For mixing, we compute the finite-window Frobenius error

$$m_h^{(F)}(s) = \left\|\mathbf{\Phi}(s+h-1, s) - \mathbf{J}\right\|_F, \qquad h \in \{1, 5, 10\}.$$

This diagnostic upper-bounds the corresponding spectral-norm error in Assumption 3. Table I reports the mean and maximum balance errors over the evaluated rounds and the mean and maximum mixing errors over the evaluated windows. The mean mixing error decreases from 0.99948 at $h = 1$ to 0.83217 at $h = 10$, whereas the maximum changes little, from 1.00818 to 1.00733. The observations support balanced aggregation and a decreasing mean finite-window mixing error. They do not establish the uniform geometric bound in Assumption 3, which remains a condition of the convergence analysis.

B. Formal Game-Theoretic Models

1) Intra-Cluster Stackelberg Game: For each kth cluster, the game Γ_k^{intra} is a two-stage dynamic game of perfect information.

Definition 1. Intra-Cluster Game The game is defined by the tuple $\Gamma_k^{\text{intra}} = \langle \mathcal{C}_k, S, \{\mathcal{A}_n\}_{n\in\mathcal{C}_k}, \{U_n\}_{n\in\mathcal{C}_k}\rangle$, where:

- Players: $\mathcal{C}_k = \{H_k\} \cup \mathcal{F}_k$ is the set of all users in the cluster, partitioned into a singleton leader H_k and a set of followers $\mathcal{F}_k$.
- Timing: The game proceeds in two stages.
  1) Stage I: The leader H_k chooses an action $A_{H_k} \in \mathcal{A}_{H_k}$.
  2) Stage II: After observing A_{H_k}, all followers $u_i \in \mathcal{F}_k$ simultaneously choose their actions $A_i \in \mathcal{A}_i$.
- Action Spaces: The action space $\mathcal{A}_i$ for any user $i \in \mathcal{C}_k$ is finite and discrete.
- Utility Functions: For each user $i \in N_k$, the utility function $U_n : \mathcal{A}_{H_k} \times \prod_{i\in\mathcal{F}_k}\mathcal{A}_i \to \mathbb{R}$ maps the complete action profile to a real-valued payoff.

2) Inter-Cluster Graph Game: The interaction among the K cluster heads is a simultaneous, non-cooperative game.

Definition 2. Inter-Cluster Game The game Γ^{inter} is defined by the tuple $\langle \mathcal{H}, \{\mathcal{A}_k\}_{k=1}^K, \{R_k\}_{k=1}^K\rangle$, where:

- Players: $\mathcal{H} = \{H_1, \ldots, H_K\}$ is the finite set of cluster heads.
- Action Spaces: Each head H_k's action space $\mathcal{A}_k$ is the power set of its physically adjacent heads, representing

TABLE I
EMPIRICAL AGGREGATION-MATRIX DIAGNOSTICS. MIXING ERRORS USE THE FROBENIUS NORM.

Diagnostic	Mean	Maximum
ϵ_{row}	7.40×10^{-17}	1.11×10^{-16}
ϵ_{col}	6.21×10^{-17}	3.42×10^{-16}
$m_1^{(F)}$	0.99948246	1.00817789
$m_5^{(F)}$	0.97556596	1.00761083
$m_{10}^{(F)}$	0.83216545	1.00733385

all possible communication link choices. $\mathcal{A}_k$ is a finite set.

- Reward Functions: The reward function $R_k : \prod_{j=1}^{K} \mathcal{A}_j \to \mathbb{R}$ for player H_k is given by:

$$R_k(\mathbf{A}_{\mathcal{H}}) = o_p \cdot \Delta\mathcal{P}_k - o_c \cdot C(A_k) + o_g \cdot \phi_k(\mathbf{A}_{\mathcal{H}}), \quad (35)$$

where $\mathbf{A}_{\mathcal{H}} = (A_1, \ldots, A_K)$ is the joint action profile and ϕ_k is the marginal contribution of H_k to the global network connectivity.

C. Proof of Equilibrium Existence

1) Proof of Stackelberg Equilibrium (SE) Existence: We use backward induction to prove the existence of a Subgame Perfect Nash Equilibrium (SPNE), which in this context is the Stackelberg Equilibrium.

a) Analysis of the Follower Subgame (Stage II).: For any fixed action A_{H_k} chosen by the leader, the followers play a simultaneous game $\Gamma_f(A_{H_k})$. A Nash Equilibrium of this subgame is an action profile $\mathbf{A}^*_{\mathcal{F}_k} = (A_i^*)_{i \in \mathcal{F}_k}$ such that no follower can improve its utility by unilaterally deviating:

$$\forall i \in \mathcal{F}_k, \quad U_i(A_i^*, \mathbf{A}^*_{-i}) \geq U_i(A_i, \mathbf{A}^*_{-i}) \quad \forall A_i \in \mathcal{A}_i. \quad (36)$$

Theorem 3. [1]
Every finite game (finite players, finite action spaces) has at least one Nash Equilibrium in mixed strategies.

Proposition 1. The follower subgame $\Gamma_f(A_{H_k})$ possesses at least one Nash Equilibrium.

Proof. The set of players $\mathcal{F}_k$ is finite, and each player's action space $\mathcal{A}_i$ is finite. By Nash's Theorem, an equilibrium exists. Let best response $BR(A_{H_k})$ be the (non-empty) set of NE profiles for the followers. □

b) Analysis of the Leader's Problem (Stage I).: The leader anticipates the followers' equilibrium response. For each leader action A_{H_k}, let σ denote a fixed equilibrium-selection rule that selects one Nash equilibrium from the non-empty follower equilibrium set $BR(A_{H_k})$, i.e.,

$$A^*_{F_k} = \sigma(BR(A_{H_k})) .$$

The selection rule is fixed ex ante and is not assumed to be either optimistic or pessimistic. Given this continuation selection, the leader's optimization problem becomes

$$\max_{A_{H_k} \in \mathcal{A}_{H_k}} U_{H_k}\left(A_{H_k}, \sigma(BR(A_{H_k}))\right) .$$

Proposition 2. An optimal action for the leader exists.

Proof. The leader's action set $\mathcal{A}_{H_k}$ is finite. The maximization problem is therefore equivalent to finding the maximum value in a finite set of real numbers $\{U_{H_k}(A) | A \in \mathcal{A}_{H_k}\}$. A maximum of a finite, non-empty set of real numbers is guaranteed to exist. □

This layer-wise selection rule is introduced only to characterize one well-defined continuation of the intra-cluster Stackelberg game. The combination of the leader's optimal action and the corresponding followers' equilibrium response constitutes a Stackelberg Equilibrium.

2) Proof of Nash Equilibrium Existence in the Graph Game: The proof for the inter-cluster game is a direct application of Nash's Theorem.

Proposition 3. The inter-cluster game Γ^{inter} possesses at least one Nash Equilibrium.

Proof. The game involves a finite set of players N_H and each player H_k has a finite action space $\mathcal{A}_k$. By Nash's Theorem, an equilibrium in mixed strategies exists. □

D. Communication Decision Complexity of Hierarchical Graph Structure

A flat decentralized graph, although conceptually simpler, becomes increasingly inefficient as the number of users grows. In a flat structure with N users, if each user evaluates candidate neighbors across the whole network, the communication decision space scales as $O(N^2)$ in the worst case, since all pairwise communication links may need to be considered. Moreover, under highly non-IID data distributions, unrestricted long-range parameter exchange may dilute local model specialization and introduce negative transfer among statistically weakly related users.

By contrast, the proposed hierarchical split reduces this burden by decomposing the global graph into an intra-cluster layer and an inter-cluster layer. Let the users be partitioned into K clusters, where cluster C_k contains $|\mathcal{C}_k|$ users. Then, the communication decision complexity in the intra-cluster layer is $O\left(\sum_{k=1}^{K} |\mathcal{C}_k|^2\right)$, since only local pairwise interactions within each cluster are considered. In addition, the inter-cluster coordination is performed only among the K cluster heads, which incurs an extra complexity of $O(K^2)$. Therefore, the total communication decision complexity becomes

$$O\left(\sum_{k=1}^{K} |\mathcal{C}_k|^2 + K^2\right).$$

Under balanced clustering with average cluster size $M = N/K$, this reduces to $O(KM^2 + K^2) = O(N^2/K + K^2)$, which is substantially smaller than $O(N^2)$ when $K \ll N$.

This hierarchy also reflects an explicit trade-off between communication complexity and model personalization. Dense global interactions in a flat graph may improve information mixing, but they also increase communication overhead and

may weaken personalized adaptation under heterogeneous non-IID data. In the proposed design, dense collaboration is restricted to local clusters, where users are geographically proximate and typically more statistically relevant, which helps preserve local personalization while reducing communication overhead. Sparse inter-cluster coordination through cluster heads then provides complementary global knowledge exchange to mitigate local overfitting without requiring all users to participate in costly long-distance communication. Therefore, the hierarchical design balances communication efficiency and personalized model adaptation in large-scale heterogeneous edge environments.

E. Existence of a Hierarchical Equilibrium in the Finite Extensive-Form Game

The proposed HierGSL framework contains several sequentially dependent decisions within each communication round. Rather than treating the intra-cluster Stackelberg game and the inter-cluster graph game as two independent games connected through equilibrium correspondences, we model the complete interaction as a single extensive-form game.

Consider communication round t and condition on the system state $\mathbf{s}(t)$ observed at the beginning of this round. Let

$$\Gamma_t^{\mathrm{H}} = \langle \mathcal{P}, \mathcal{H}, \{\mathcal{A}(h)\}_{h\in\mathcal{H}}, \{\mathcal{I}_p\}_{p\in\mathcal{P}}, \{U_p\}_{p\in\mathcal{P}} \rangle$$

denote the resulting hierarchical extensive-form game, where $\mathcal{P}$ is the player set, $\mathcal{H}$ is the set of feasible histories, $\mathcal{A}(h)$ is the feasible action set following history h, $\mathcal{I}_p$ is the collection of information sets of player p, and U_p is the expected terminal utility of player p.

The game proceeds through the following stages.

Stage 1: Incentive-intensity commitment. At the beginning of communication round t, each cluster head H_k selects and publicly announces an incentive intensity

$$\eta_k(t) \in \mathcal{E}_k,$$

where $\mathcal{E}_k$ denotes the admissible set of incentive intensities. The announced $\eta_k(t)$ constitutes the leader's strategic action in the intra-cluster Stackelberg interaction and remains fixed throughout the current intra-cluster decision stage.

Stage 2: Intra-cluster follower decisions. After observing the incentive intensity $\eta_k(t)$ announced by its cluster head and its available local information, each follower $u_i \in \mathcal{C}_k$ selects

$$A_i(t) = \big(\mathbf{a}_i(t), e_i(t)\big) \in \mathcal{A}_i,$$

where $\mathbf{a}_i(t)$ is a binary intra-cluster neighbor-selection vector and $e_i(t)$ is selected from a bounded set of integer-valued local-training epochs. Hence, $\mathcal{A}_i$ is finite. Followers within the same cluster may make their decisions simultaneously, which is represented through the corresponding information sets of the extensive-form game.

Stage 3: Incentive-payment realization. After the follower actions have been executed, the resulting scheduling utility and energy consumption are observed. The incentive payment is then mechanically settled according to the incentive intensity announced in Stage 1:

$$I_i(t) = \eta_k(t)\frac{P_i\Big(\theta_i^{e_i(t),\mathbf{a}_i(t)}\Big)}{E_i^c(e_i(t)) + E_i^{\mathrm{intra,tr}}(\mathbf{a}_i(t)) + \epsilon}. \tag{37}$$

Importantly, neither $\eta_k(t)$ nor the settlement rule is changed at this stage. Therefore, $I_i(t)$ is an outcome-contingent payment generated from the previously announced leader action and the realized follower action, rather than an additional strategic choice of the cluster head.

Stage 4: Inter-cluster graph decisions. After observing the intra-cluster outcomes, each cluster head H_k selects its inter-cluster graph action

$$\mathbf{b}_k(t) \in \mathcal{B}_k,$$

where $\mathcal{B}_k$ is the finite set of feasible binary inter-cluster neighbor-selection vectors. The cluster heads may select these actions simultaneously, thereby inducing a finite strategic-form continuation subgame.

For every terminal history, each player is assigned a single scalar payoff. For a follower u_i, the terminal payoff is its intra-cluster utility defined in Section IV. For a cluster head H_k, the terminal payoff incorporates both its intra-cluster leader utility and its subsequent inter-cluster reward, i.e.,

$$U_{H_k}^{\mathrm{H}} = U_{H_k}^{\mathrm{intra}} + R_{H_k}^{\mathrm{inter}},$$

where the weighting parameters contained in the two component utilities determine the corresponding trade-offs. Any stochasticity arising from local training or communication is incorporated into the expected utilities conditional on the current state and the selected action history.

Assumption E1 (Finite strategic choices). The player set is finite, and

$$|\mathcal{E}_k| < \infty, \qquad |\mathcal{A}_i| < \infty, \qquad |\mathcal{B}_k| < \infty,$$

for all cluster heads and followers. In particular, for the finite-game equilibrium analysis, the admissible incentive intensities are selected from a finite codebook

$$\mathcal{E}_k = \left\{\eta_k^{(1)}, \ldots, \eta_k^{(M_k)}\right\}.$$

The follower action space is finite because the local-training epochs are bounded integers and the intra-cluster neighbor-selection variables are binary, while the inter-cluster graph actions are finite binary vectors.

The finite incentive-intensity codebook is introduced solely for the equilibrium-existence analysis. In the experimental implementation, the MADDPG leader policy instead selects incentive intensities from a bounded continuous interval. The follower decisions remain integer-valued for local epochs and binary for neighbor selection. Therefore, the finite-codebook assumption is not imposed on the experimental leader policy, and the equilibrium-existence result applies only to the finite-action game defined in this appendix.

Assumption E2 (Well-defined utilities). For every terminal action history, the expected utility of each player is finite and well defined.

Assumption E3 (Perfect recall and stage-wise observability). Each follower observes the incentive intensity announced by its cluster head before selecting its action, and the cluster heads observe the realized intra-cluster outcomes before selecting their inter-cluster graph actions. Every player remembers its own previous observations and actions. Therefore, the extensive-form game has perfect recall and the continuation games considered below are well defined.

Definition 4 (Hierarchically Coupled Equilibrium). A behavioral strategy profile

$$\boldsymbol{\sigma}^\star = \left(\sigma_p^\star\right)_{p\in\mathcal{P}}$$

is called a hierarchically coupled equilibrium (HCE) if it is a subgame-perfect equilibrium of Γ_t^{H}; that is, the restriction of $\boldsymbol{\sigma}^\star$ to every subgame of Γ_t^{H}, including the entire game itself, constitutes a Nash equilibrium of that subgame.

The terminal action distribution induced by $\boldsymbol{\sigma}^\star$ is referred to as the HCE outcome.

The terminal action distribution induced by $\boldsymbol{\sigma}^\star$ is referred to as the HCE outcome.

Theorem 4 (Existence of a Hierarchically Coupled Equilibrium)**.** Under Assumptions E1-E3, the finite hierarchical extensive-form game Γ_t^{H} admits at least one mixed-strategy subgame-perfect equilibrium. Because the game has perfect recall, this equilibrium admits a realization-equivalent behavioral-strategy representation. Therefore, the proposed hierarchical interaction admits at least one HCE.

Proof. Under Assumptions E1-E3, Γ_t^{H} is a finite extensive-form game with well-defined expected utilities and perfect recall.

Every proper subgame of Γ_t^{H} is itself a finite extensive-form game. Starting from the terminal proper subgames, the mixed extension of each corresponding finite continuation game admits at least one Nash equilibrium. Fix one such equilibrium in each terminal proper subgame and replace that subgame by its induced continuation payoff. Repeating this argument recursively over the finite subgame structure leaves a finite root game with continuation payoffs fixed by the equilibria already selected. We then select a mixed-strategy Nash equilibrium of this remaining root game, which exists by Nash's theorem applied to its finite strategic-form representation. This final step covers the cluster heads' initial incentive-intensity decisions $\{\eta_k(t)\}_{k=1}^K$ and rules out profitable unilateral deviations at that stage under the selected continuation strategies. Combining the root-game equilibrium with the selected continuation equilibria yields a complete strategy profile whose restriction to every subgame, including Γ_t^{H} itself, is a Nash equilibrium.

Hence, Γ_t^{H} admits at least one mixed-strategy subgame-perfect equilibrium.

Finally, because Γ_t^{H} has perfect recall under Assumption E3, the mixed-strategy equilibrium admits a realization-equivalent behavioral-strategy representation. According to Definition 4, this behavioral strategy profile constitutes an HCE. □

Remark 1 (Follower-equilibrium selection). A follower continuation game may admit multiple mixed-strategy Nash equilibria. The existence theorem does not impose an optimistic or pessimistic follower response. Instead, a complete subgame-perfect equilibrium specifies one equilibrium continuation strategy following every feasible incentive-intensity history, including histories that are not reached on the equilibrium path. An optimistic or pessimistic Stackelberg refinement would require an additional equilibrium-selection or tie-breaking rule and is not assumed here.

Remark 2 (Role of the incentive intensity and payment). The strategic leader action in the intra-cluster extensive-form interaction is the incentive intensity $\eta_k(t)$ announced before the followers act. Once announced, $\eta_k(t)$ remains fixed during the current intra-cluster interaction. The realized payment $I_i(t)$ is generated only after the follower action according to Eq. (37). Hence,

$$\eta_k(t) \longrightarrow \big(\mathbf{a}_i(t), e_i(t)\big) \longrightarrow I_i(t),$$

which preserves the leader-first timing required by the Stackelberg formulation.

Remark 3 (Scope of the result). Theorem 4 establishes the existence of an equilibrium for the finite hierarchical game defined above. In particular, the result relies on the finite incentive-intensity codebook in Assumption E1 and therefore applies to the discretized strategic game used for the equilibrium-existence analysis.

This equilibrium-existence result is conceptually separated from the continuous-function-approximation dynamics of the MADDPG and IDQN solvers. It neither establishes equilibrium existence for the learning dynamics themselves nor guarantees that MADDPG or IDQN converges to the HCE.

F. Optimality Gap of the Two-Stage Decomposition

The proposed HierGSL framework solves the coupled optimization problem in a sequential two-stage manner. To characterize the approximation error induced by this decomposition, we analyze the deviation between the sequentially updated solution and the joint optimum, followed [2].

Let the system-level objective be

$$F(x, y),$$

where $x \in \mathcal{X}$ denotes the intra-cluster decision variables and $y \in \mathcal{Y}$ denotes the inter-cluster decision variables. Assume that $(x^\star, y^\star)$ is a global maximizer of

$$\max_{x\in\mathcal{X},\, y\in\mathcal{Y}} F(x, y).$$

Since $(x^\star, y^\star)$ is globally optimal, it is also blockwise optimal, i.e.,

$$x^\star \in \arg\max_{x\in\mathcal{X}} F(x, y^\star), \qquad y^\star \in \arg\max_{y\in\mathcal{Y}} F(x^\star, y).$$

For any given y, define the intra-cluster best-response mapping

$$T_x(y) := \arg\max_{x\in\mathcal{X}} F(x, y),$$

and for any given x, define the inter-cluster best-response mapping

$$T_y(x) := \arg\max_{y\in\mathcal{Y}} F(x,y).$$

Under the two-stage decomposition, starting from an input y^0, the sequential update is

$$x^d = T_x(y^0), \qquad y^d = T_y(x^d).$$

We first establish the sensitivity of the two blockwise maximizers.

Proposition 4. Suppose that
(i) $F(x,y)$ is continuously differentiable;
(ii) for every fixed y, $F(x,y)$ is μ_x-strongly concave in x;
(iii) for every fixed x, $F(x,y)$ is μ_y-strongly concave in y;
(iv) the cross-layer gradients satisfy

$$\|\nabla_x F(x,y_1) - \nabla_x F(x,y_2)\| \le L_{xy}\|y_1 - y_2\|,$$
$$\|\nabla_y F(x_1,y) - \nabla_y F(x_2,y)\| \le L_{yx}\|x_1 - x_2\|.$$

Then the best-response mappings satisfy

$$\|T_x(y_1) - T_x(y_2)\| \le \frac{L_{xy}}{\mu_x}\|y_1 - y_2\|,$$

and

$$\|T_y(x_1) - T_y(x_2)\| \le \frac{L_{yx}}{\mu_y}\|x_1 - x_2\|.$$

Proof. Let $x_i = T_x(y_i)$ for $i = 1, 2$. Since x_i maximizes $F(x, y_i)$, the first-order optimality condition gives

$$\nabla_x F(x_1,y_1) = 0, \qquad \nabla_x F(x_2,y_2) = 0.$$

Hence,

$$\nabla_x F(x_1,y_1) - \nabla_x F(x_2,y_1) = \nabla_x F(x_2,y_2) - \nabla_x F(x_2,y_1).$$

Taking norms and using the μ_x-strong concavity of $F(\cdot, y_1)$ yields

$$\mu_x\|x_1 - x_2\| \le \|\nabla_x F(x_1,y_1) - \nabla_x F(x_2,y_1)\|.$$

By the cross-layer Lipschitz condition,

$$\|\nabla_x F(x_2,y_2) - \nabla_x F(x_2,y_1)\| \le L_{xy}\|y_2 - y_1\|.$$

Combining the above inequalities gives

$$\|x_1 - x_2\| \le \frac{L_{xy}}{\mu_x}\|y_1 - y_2\|.$$

Thus,

$$\|T_x(y_1) - T_x(y_2)\| \le \frac{L_{xy}}{\mu_x}\|y_1 - y_2\|.$$

The proof for $T_y(\cdot)$ is analogous:

$$\|T_y(x_1) - T_y(x_2)\| \le \frac{L_{yx}}{\mu_y}\|x_1 - x_2\|.$$

■

Using Proposition B.1, we can now bound the deviation induced by one sequential two-stage decomposition.

Proposition 5. Under the assumptions of Proposition B.1, the two-stage update satisfies

$$\|x^d - x^\star\| \le \frac{L_{xy}}{\mu_x}\|y^0 - y^\star\|,$$

and

$$\|y^d - y^\star\| \le \frac{L_{yx}L_{xy}}{\mu_y\mu_x}\|y^0 - y^\star\|.$$

Proof. Since $x^\star = T_x(y^\star)$ and $x^d = T_x(y^0)$, Proposition B.1 implies

$$\|x^d - x^\star\| = \|T_x(y^0) - T_x(y^\star)\| \le \frac{L_{xy}}{\mu_x}\|y^0 - y^\star\|.$$

Similarly, since $y^\star = T_y(x^\star)$ and $y^d = T_y(x^d)$,

$$\|y^d - y^\star\| = \|T_y(x^d) - T_y(x^\star)\| \le \frac{L_{yx}}{\mu_y}\|x^d - x^\star\|.$$

Substituting the first bound into the second gives

$$\|y^d - y^\star\| \le \frac{L_{yx}L_{xy}}{\mu_y\mu_x}\|y^0 - y^\star\|.$$

■

To convert the variable deviation into an objective-value gap, we further assume that F is jointly L_F-smooth, i.e.,

$$\|\nabla F(z_1) - \nabla F(z_2)\| \le L_F\|z_1 - z_2\|, \qquad z := (x,y).$$

Then, by the standard smoothness inequality and the first-order optimality of $(x^\star, y^\star)$,

$$F(x^\star,y^\star) - F(x^d,y^d) \le \frac{L_F}{2}\Big(\|x^d - x^\star\|^2 + \|y^d - y^\star\|^2\Big).$$

Combining this with Proposition B.2 yields the following theorem.

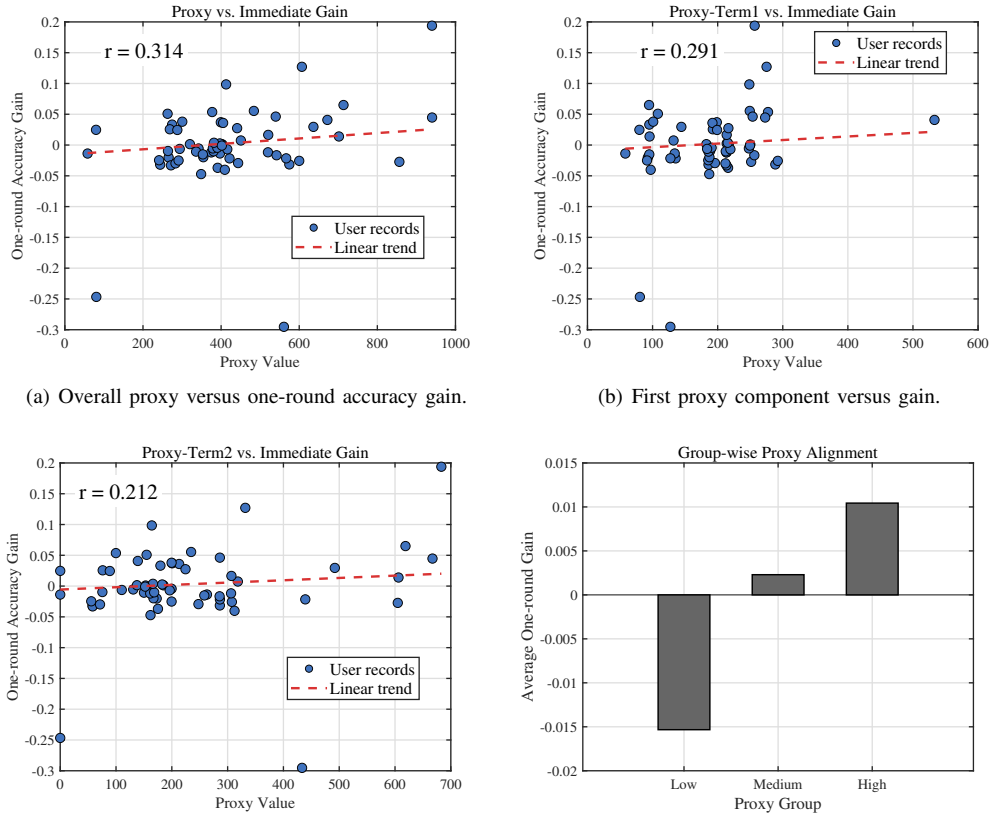
Theorem 5. Suppose that the assumptions of Proposition B.1 hold and that F is jointly L_F-smooth. Then the decomposition-induced optimality gap satisfies

$$\Delta := F(x^\star,y^\star) - F(x^d,y^d) \le \frac{L_F}{2}\left[\left(\frac{L_{xy}}{\mu_x}\right)^2 + \left(\frac{L_{yx}L_{xy}}{\mu_y\mu_x}\right)^2\right]\|y^0 - y^\star\|^2.$$

This bound shows that the approximation error of the sequential decomposition is controlled by the cross-layer coupling constants L_{xy} and L_{yx}. In particular, when the interaction between the two layers becomes weak, the best-response mappings become less sensitive to each other, and the decomposition gap correspondingly decreases.

G. Empirical Association Between the Heuristic Utility in Eq. (10) and One-Round Local Performance Change

To further examine the scheduling relevance of Eq. (10), we conduct an empirical study of its association with the realized one-round change in each user's local task performance. This analysis does not treat Eq. (10) as a direct measure or an accurate predictor of personalization performance. Instead, it examines whether the proposed heuristic utility contains partial



(a) Overall proxy versus one-round accuracy gain.

(b) First proxy component versus gain.

(c) Second proxy component versus gain.

(d) Average gain under low-, medium-, and high-proxy groups.

Fig. 1. Empirical alignment between Eq. (10) and realized personalized task gain.

ranking information that can support relative prioritization among candidate training and communication actions.

Specifically, during training, we record, for each user and each communication round, the utility value defined in Eq. (10) and the local test-accuracy change observed over the same round after local updating and decentralized aggregation. The one-round accuracy change is calculated as

$$\Delta A_i^{(t)} = A_i^{(t+1)} - A_i^{(t)}, \tag{38}$$

where $A_i^{(t)}$ and $A_i^{(t+1)}$ denote the test accuracies of user u_i before and after the corresponding communication round, respectively. This quantity reflects the realized short-term change in local task performance. It does not isolate the causal contribution of the scheduling utility and should not be interpreted as a complete measure of long-term personalization performance.

The dataset and data-partition details are provided in Appendix H. This association analysis includes 20 distinct users sampled at communication rounds 50, 100, and 150, with 20 user-round observations per sampled round and 60 observations in total. All 60 records are included in the analysis.

Based on the recorded samples, we conduct both sample-level and group-level analyses. At the sample level, we examine the association between the heuristic utility and the realized accuracy change. At the group level, all samples are partitioned into low-, medium-, and high-utility groups according to the lower, middle, and upper tertiles, and the average accuracy change within each group is compared. The group-level analysis is included because one-round accuracy changes are also influenced by optimization noise, the local convergence stage, stochastic mini-batch sampling, and the compatibility of the received model updates. Therefore, the grouping results are used as a descriptive robustness analysis rather than as evidence of a deterministic relationship.

The sample-level statistic is Pearson's correlation coefficient, computed directly over all 60 pooled user-round observations with equal weight per observation, without averaging round-specific correlations. The group-level tertile cut points are determined from the pooled overall utility values, not separately within each round. Each of the low-, medium-, and high-utility groups contains 20 user-round observations.

The records are repeated observations of the same 20 users. Observations from the same user share its local data and evolving model state and are dependent across rounds. Thus, the 60 records are not treated as 60 independent experimental replications. The pooled correlations and group means are reported descriptively, without confidence intervals, standard errors, or p-values.

Fig. 1(a)-(c) illustrate the associations between the heuristic utility and the realized one-round accuracy change. The sample-level correlation coefficients are positive but modest, and the observations exhibit substantial dispersion. These re-

sults indicate that Eq. (10) captures only part of the factors associated with short-term local performance changes. The local-drift component and the neighbor-discrepancy component exhibit similarly weak-to-moderate positive tendencies. Thus, both components contain some information relevant to scheduling, but neither component should be interpreted as an independent or direct indicator of personalization performance.

Fig. 1(d) presents the corresponding group-level results. The average accuracy changes exhibit a monotonic ordering across the three utility groups: the low-utility group has a negative average change, the medium-utility group has a near-zero or slightly positive average change, and the high-utility group achieves the largest positive average change. This aggregate ordering is consistent with the intended use of Eq. (10) for coarse relative prioritization. However, it does not imply accurate sample-level prediction or a causal relationship between the utility value and the subsequent performance change.

Taken together, the results provide limited but useful empirical support for interpreting Eq. (10) as a heuristic personalization-aware scheduling utility. The metric provides partial ranking information for training, communication, and resource-allocation decisions, for which relative prioritization is more relevant than exact performance prediction. The actual personalization performance of HierGSL is evaluated separately using user-specific test accuracy or MSE over the complete training process.

TABLE II
COMPARISON OF SCHEDULING UTILITIES. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE RANDOM SEEDS.

Metric	Eq. (10)	Validation gain
Final accuracy (%)	68.92 ± 1.14	67.38 ± 1.04
Training energy (10^8 model units)	3.358 ± 0.254	3.380 ± 0.262
Communication energy (10^{11} model units)	2.284 ± 0.989	2.322 ± 1.017
Validation time (s)	0	480.49 ± 3.78
Validation sample-forward evaluations (10^6)	0	4.00 ± 0.00

To complement the correlation analysis, we compare Eq. (10) with a validation-gain-based scheduling utility that uses measured validation improvement as the task-related scheduling signal. The HierGSL framework and experimental settings are kept unchanged, while the scheduling utility is replaced. Both variants are evaluated using three random seeds. Table II reports the mean and standard deviation of final accuracy, modeled training and communication energy, and additional validation evaluation overhead.

Eq. (10) achieves a final accuracy of $68.92 \pm 1.14\%$, compared with $67.38 \pm 1.04\%$ for the validation-gain utility, giving a mean difference of 1.54 percentage points. The two variants incur comparable modeled training and communication energy. The validation-gain utility additionally requires 480.49 ± 3.78 seconds of validation evaluation and 4.00×10^6 sample-forward evaluations per run. Eq. (10) avoids these additional validation evaluations.

This comparison supports the practical use of Eq. (10) as a scheduling utility in the evaluated setting: it achieves a higher mean final accuracy at comparable modeled energy cost without additional validation evaluation. The result concerns the performance of the resulting scheduling policy and does not imply that increasing either parameter-space norm always improves generalization.

H. Details of Experimental Setup

Non-IID Data Partition and Heterogeneous Device Initialization: To improve reproducibility, we further clarify the non-IID data partition and heterogeneous device initialization used in the experiments.

For the CIFAR-10 dataset, we adopt a Dirichlet-based non-IID partition strategy combined with region-dependent data-size allocation to jointly model statistical and quantity heterogeneity. Specifically, the whole training set is first sorted by class labels and divided into 200 equal-size shards, where each shard contains samples predominantly from a single class. This shard-based organization facilitates controlled data assignment at the class level.

Then, the total number of samples assigned to each user depends on its regional type rather than being uniformly distributed. The data-allocation weights are set as 3.0, 2.5, 2.0, 1.5, 1.0, and 0.5 for users located in *Business District*, *Urban*, *Suburban*, *County*, *Rural*, and *Mountain* areas, respectively. After normalization, each user u_i is assigned a target local dataset size $|D_i|$, which introduces quantity heterogeneity across users.

Next, for each user u_i, its class distribution is generated by sampling from a Dirichlet distribution:

$$\boldsymbol{\pi}_i \sim \mathrm{Dir}(\alpha_i \mathbf{1}),$$

where $\boldsymbol{\pi}_i$ denotes the class proportion vector of user u_i, and the concentration parameter α_i is set according to the regional type, taking values from $\{1.0, 0.8, 0.6, 0.4, 0.2, 0.1\}$ for *Business District*, *Urban*, *Suburban*, *County*, *Rural*, and *Mountain* users, respectively. A larger α_i leads to a more balanced local class distribution, while a smaller α_i results in a more skewed and heterogeneous dataset.

Finally, based on the sampled class proportions $\boldsymbol{\pi}_i$ and the target dataset size $|D_i|$, shards are allocated to each user from class-specific shard pools. Therefore, the resulting local datasets differ in both total size and class composition, yielding both quantity heterogeneity and statistical heterogeneity across users.

Meanwhile, user heterogeneity is modeled through region-dependent normalized computation and communication coefficients. The normalized computation coefficients are initialized as 1.2, 1.0, 0.8, 0.6, 0.4, and 0.3 for *Business District*, *Urban*, *Suburban*, *County*, *Rural*, and *Mountain* users, respectively, which serve as normalized CPU-frequency indicators. Accordingly, the local training energy of user u_i is modeled as

$$E_i^{\mathrm{train}} = \alpha \chi_i |D_i| e_i N_i^{\mathrm{batch}},$$

where χ_i denotes the computation coefficient, $|D_i|$ denotes local dataset size, e_i denotes local training epochs, and N_i^{batch} denotes the number of mini-batches.

Similarly, communication heterogeneity is characterized by normalized communication-capacity coefficients α initialized as 1.5, 1.2, 1.0, 0.8, 0.6, and 0.5 for the six regional user types, respectively. The communication energy for model exchange between users is computed as

$$E_{ij}^{\mathrm{comm}} = \beta \frac{S_{\mathrm{model}} d_{ij}}{c_i},$$

where S_{model} denotes model size, d_{ij} denotes geographical distance, and c_i denotes the communication-capacity coefficient of user u_i. Here, the proportional constant β absorbs transmission-power-related physical factors.

Unless otherwise specified, the experiments are initialized with 100 users and 10 clusters. Each user adopts a ResNet-18 local model trained by SGD optimizer with learning rate 0.001, mini-batch size 64, and local epoch decision range from 1 to 10. The epoch range 1-10 applies to active users; a user that cannot afford one local epoch takes the no-operation action specified in Section IV-B. The user clustering is initialized by K-means with random seed 42.

REFERENCES

[1] J. F. Nash, "Non-cooperative games," in *Found. Price Theory Vol. 4*. Routledge, 2024, pp. 329–340.

[2] K. Du, C.-C. Ruan *et al.*, "On the convergence of a randomized block coordinate descent algorithm for a matrix least squares problem," *Appl. Math. Lett.*, vol. 124, p. 107689, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0893965921003682